## Session 2018-2019

**1.**

**a. Define the following terminologies: i) System ii) Model iii) Simulation iv) Analytical solution. Demonstrate different ways in which a system might be studied.**

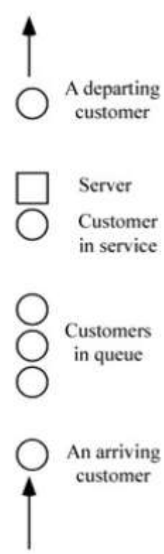
- ***System:*** A system is defined to be a collection of entities, e.g., people or machines, that act and interact together toward the accomplishment of some logical end.

  Two types of system:
  - ✓ **Discrete system:** The system in which the state changes abruptly at discrete points in time are called discrete systems.
    *Example:* A bank is an example of a discrete system.
  - ✓ **Continuous system:** System in which the state of the system changes continuously with time are called continuous system.
    *Example:* An airplane moving through the air is an example of a continuous system.
- **Model:** A model is a simplified representation of a real-world system or process used to understand or predict how it behaves.
  *Example:* A toy car can be a model of a real car to show how it moves.
  - ✓ **Physical Model:** A physical model is a tangible, small-scale version of something larger. Like a model airplane that looks and behaves like a real airplane.
  - ✓ **Mathematical Model:** A mathematical model uses equations and formulas to represent how a system works. A simple equation like $d=r \times t$ can be used to represent the relationship between distance, rate (or speed), and time.
- **Simulation:** Simulation is the imitation of reality. *Example:* Flight simulator.
- **Analytical solution**: An analytical solution is a precise, exact answer to a problem derived using mathematical methods.
  *Example:* Solving the equation $x+2=5$ to get $x=3$.

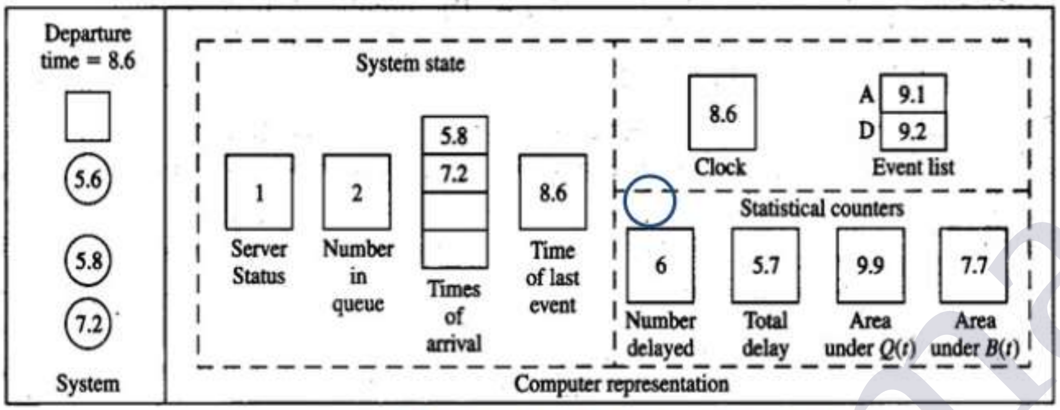
**b. What are the steps of simulation for a single-server queueing system? Describe the problem statement and intuitive explanation of a single-server queueing system.**

**1. Problem statement:**

- Recall single-server queuing model
- Assume interarrival times are independent and identically distributed (IID) random variables
- Assume service times are IID, and are independent of interarrival times
- Queue discipline is FIFO
- Start empty and idle at time 0
- First customer arrives after an interarrival time, not at time 0
- Stopping rule: When *n*th customer has completed delay in queue (i.e., *enters* service) … *n* will be specified as input



2. **Intuitive explanation.** Try to include at least first two process and last process in exam.




Interarrival times: ~~0.4~~, ~~1.2~~, ~~0.5~~, ~~1.7~~, ~~0.2~~, ~~1.6~~, ~~0.2~~, ~~1.4~~, ~~1.9~~, ...
Service times: ~~2.0~~, ~~0.7~~, ~~0.2~~, ~~1.1~~, ~~3.7~~, ~~0.6~~, ...

**Final output performance measures:**

**Average delay in queue = 5.7/6 = 0.95 min./cust.**
**Time-average number in queue = 9.9/8.6 = 1.15 custs.**
**Server utilization = 7.7/8.6 = 0.90 (dimensionless)**

3. **Program organization and logic. (First two step for second part question answer)**

- How to "draw" (or generate) an observation (*variate*) from an exponential distribution?
- Proposal:
  - Assume a perfect *random-number generator* that generates IID variates from a continuous uniform distribution on [0, 1] ...
  - Algorithm:
    1. Generate a random number $U$
    2. Return $X = -\beta \ln U$
  - Proof that algorithm is correct

$$
\begin{aligned}
P(\text{generated } X \le x) &= P(-\beta \ln U \le x) \\
&= P(\ln U \ge -x/\beta) \\
&= P(U \ge e^{-x/\beta}) \\
&= P(e^{-x/\beta} \le U \le 1) \\
&= 1 - e^{-x/\beta}
\end{aligned}
$$

4. **C Program:** C program for simulating an M/M/1 queue, emphasizing the use of clear variable names and function prototyping. It also notes that floating-point inaccuracies in different compilers can affect simulation results, especially when events are close in time.

5. **Simulation Output and Discussion:** The results of an M/M/1 queue simulation, highlighting that the output numbers, such as average delay, number in the queue, and server utilization, are influenced by random number generation and input parameters. It discusses the challenges of choosing a stopping rule for accurate steady-state estimation and suggests focusing on the delay in the queue rather than the total waiting time for efficiency in estimating system performance.
6. **Alternative stopping rules**: The text describes modifying a queueing simulation to stop after a fixed amount of time, such as 8 hours, by introducing a dummy "end-simulation" event. This event, scheduled to occur at the specified time, ensures that the simulation stops when the event's time is reached. The changes involve adjusting the program's external definitions, main function, and initialization and reporting functions to handle this fixed run length and manage the end of the simulation properly.
7. **Determining the Events and Variables:** The text discusses the event-graph method for modeling complex systems, where events are represented as nodes connected by arrows that show how they affect each other. It explains that this technique helps simplify the model by eliminating unnecessary events and ensuring correct initialization. The method is useful for managing and visualizing event sequences in simulations, especially in complex systems, and can also help detect errors and predict computational demands

**2.**

a. **What is random variable? Compute variance and covariance based on a set of random variates. Why stochastic process is used to simulate the output data? Illustrate correlation function of the process of delays in queue D1, D2, ...... for the M/M/1 queue.**

EXAMPLE 4.3. Consider the experiment of rolling a pair of dice. Then

$$S = \{(1, 1), (1, 2), \ldots, (6, 6)\}$$

where $(i, j)$ means that $i$ and $j$ appeared on the first and second die, respectively. If $X$ is the random variable corresponding to the sum of the two dice, then $X$ assigns the value 7 to the outcome (4, 3).

**Variance:**

EXAMPLE 4.16. For the uniform random variable on [0, 1] in Example 4.7, the variance is computed as follows:

$$E(X^2) = \int_0^1 x^2 f(x)\, dx = \int_0^1 x^2\, dx = \frac{1}{3}$$

$$\text{Var}(X) = E(X^2) - \mu^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}$$

**EXAMPLE 4.17.** For the jointly continuous random variables $X$ and $Y$ in Example 4.11, the covariance is computed as

$$\begin{aligned}
E(XY) &= \int_0^1 \int_0^{1-x} xy\, f(x, y)\, dy\, dx \\
&= \int_0^1 x^2 \left( \int_0^{1-x} 24y^2\, dy \right) dx \\
&= \int_0^1 8x^2(1-x)^3\, dx \\
&= \frac{2}{15}
\end{aligned}$$

$$E(X) = \int_0^1 x f_X(x)\, dx = \int_0^1 12x^2(1-x)^2\, dx = \frac{2}{5}$$

$$E(Y) = \int_0^1 y f_Y(y)\, dy = \int_0^1 12y^2(1-y)^2\, dy = \frac{2}{5}$$

Therefore,

$$\begin{aligned}
\operatorname{Cov}(X, Y) &= E(XY) - E(X)E(Y) \\
&= \frac{2}{15} - \left(\frac{2}{5}\right)\left(\frac{2}{5}\right) \\
&= -\frac{2}{75}
\end{aligned}$$

A **stochastic process** is used to simulate situations where outcomes are uncertain or random. It helps predict possible results when we can't know for sure what will happen, but we can estimate based on probabilities.

**Example:** Imagine you're trying to guess how many customers visit a shop each day. Some days there are 20, other days 50, and it changes randomly. A stochastic process helps simulate these changes, giving you a range of possible outcomes instead of just one fixed number.

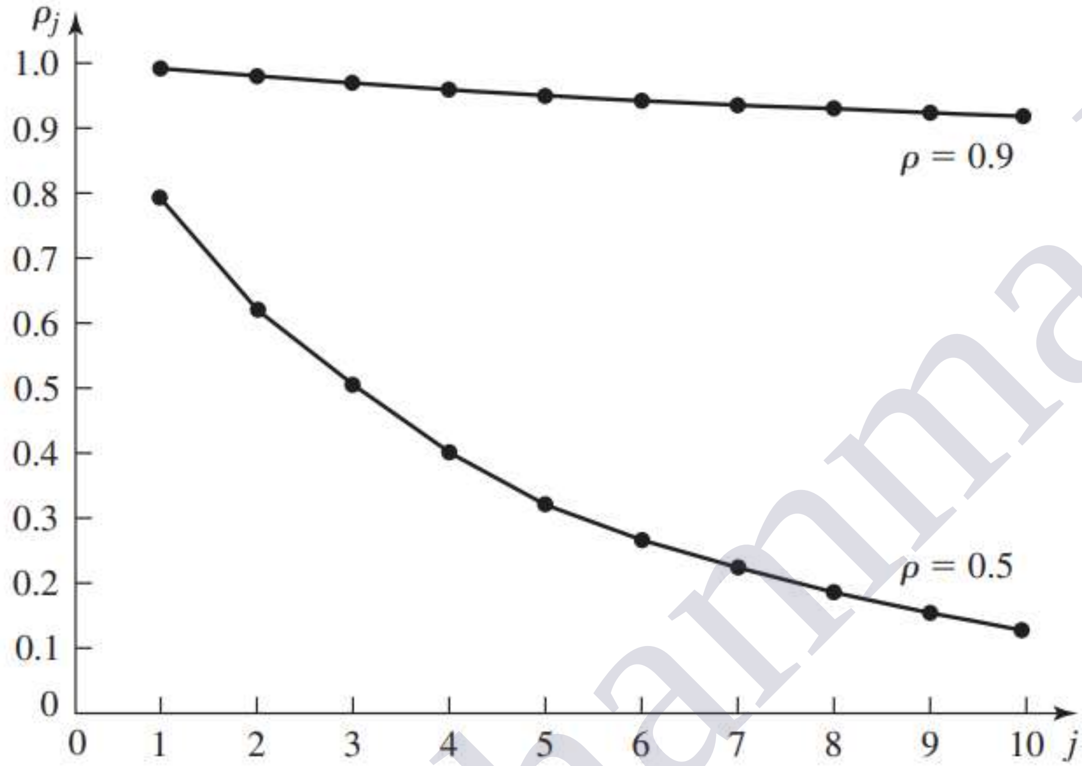



**FIGURE 4.10**
Correlation function $\rho_j$ of the process $D_1, D_2, \ldots$ for the $M/M/1$ queue.

**EXAMPLE 4.22.** Consider the output process $D_1, D_2, \ldots$ for a covariance-stationary (see App. 4A for a discussion of this technical detail) $M/M/1$ queue with $\rho = \lambda/\omega < 1$ (recall that $\lambda$ is the arrival rate and $\omega$ is the service rate). From results in Daley (1968), one can compute $\rho_j$, which we plot in Fig. 4.10 for $\rho = 0.5$ and 0.9. (Do not confuse $\rho_j$ and $\rho$.) Note that the correlations $\rho_j$ are positive and monotonically decrease to zero as $j$ increases. In particular, $\rho_1 = 0.99$ for $\rho = 0.9$ and $\rho_1 = 0.78$ for $\rho = 0.5$. Furthermore, the convergence of $\rho_j$ to zero is considerably slower for $\rho = 0.9$; in fact, $\rho_{50}$ is (amazingly) 0.69. (In general, our experience indicates that output processes for queueing systems are positively correlated.)

**b. Explain hypothesis testing including null and alternative hypothesis for mean. What is the difference type I error and type II error? Explain and illustrate the strong law of large number.**

When one performs a hypothesis test, two types of errors can be made. If one rejects $H_0$ when in fact it is true, this is called a Type I error. The probability of Type I error is equal to the level a and is thus under the experimenter's control. If one fails to reject $H_0$ when it is false, this is called a Type II error. For a fixed level a and sample size n, the probability of a Type II error, which we denote by b, depends on what is actually true (other than $H_0$: $\mu = \mu_0$), and is usually unknown. We call $\delta$=1-$\beta$ the power of the test, and it is equal to the probability of rejecting H0 when it is false. There are four different situations that can occur when one tests the null hypothesis $H_0$ against the alternative hypothesis $H_1$, and these are delineated in Table 4.2 along with their probabilities of occurrence.

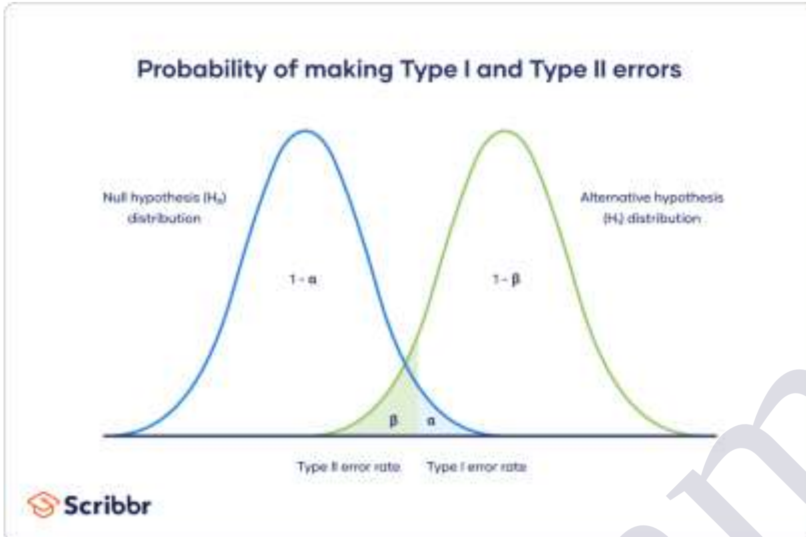
**TABLE 4.2**

**Hypothesis-testing situations and their corresponding probabilities of occurrence**

| Outcome \ $H_0$ | True | False |
|---|---|---|
| Reject | $\alpha$ | $\delta = 1 - \beta$ |
| Fail to reject | $1 - \alpha$ | $\beta$ |

**The key differences between Type I and Type II errors:**

| Aspect | Type I Error (False Positive) | Type II Error (False Negative) |
|---|---|---|
| **Definition** | Rejecting a true null hypothesis | Failing to reject a false null hypothesis |
| **Interpretation** | Detecting an effect that doesn't exist | Missing an effect that actually exists |
| **Common Example** | Concluding a treatment works when it doesn't | Concluding a treatment doesn't work when it does |
| **Probability Notation** | $\alpha$ (alpha) | $\beta$ (beta) |
| **Consequence** | Incorrectly claiming significance | Missing a true significant finding |
| **Relation to Power** | Lower $\alpha$ reduces chance of Type I error, but increases risk of Type II error | Higher power (1 - $\beta$) reduces the chance of Type II error |




The second most important result in probability theory (after the central limit theorem) is arguably the strong law of large numbers. Let $X_1, X_2, \ldots, X_n$ be IID random variables with finite mean $\mu$. Then the *strong law of large numbers* is as follows [see Chung (1974, p. 126) for a proof].

**THEOREM 4.2.** $\bar{X}(n) \rightarrow \mu$ w.p. 1 as $n \rightarrow \infty$.

The theorem says, in effect, that if one performs an infinite number of experiments, each resulting in an $\bar{X}(n)$, and $n$ is sufficiently large, then $\bar{X}(n)$ will be arbitrarily close to $\mu$ for almost all the experiments.

**EXAMPLE 4.31.** Suppose that $X_1, X_2, \ldots$ are IID normal random variables with $\mu = 1$ and $\sigma^2 = 0.01$. Figure 4.18 plots the values of $\bar{X}(n)$ for various $n$ that resulted from sampling from this distribution. Note that $\bar{X}(n)$ differed from $\mu$ by less than 1 percent for $n \geq 28$.

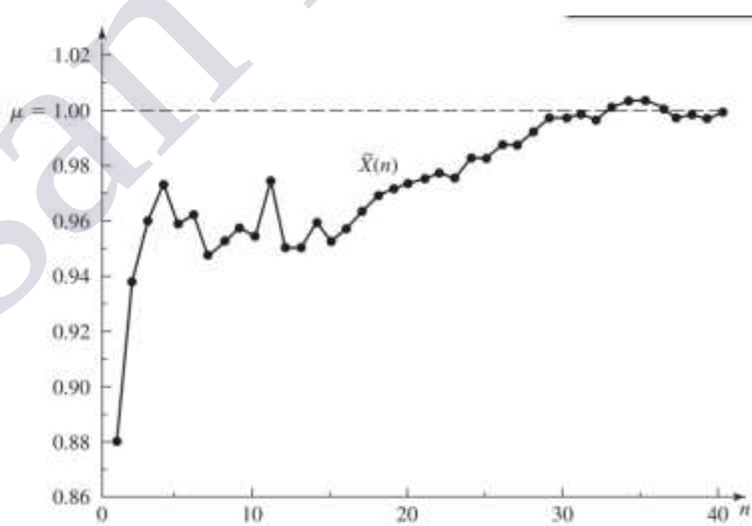



**FIGURE 4.18**
$\bar{X}(n)$ for various values of $n$ when the $X_i$'s are normal random variables with $\mu = 1$ and $\sigma^2 = 0.01$.